Algebra II with Trigonometry Chapter 11.1

Olympiad Problems (Gemini)

Saved tutoring response with MathJax rendering. Original PDF file: 4652951_alg-2trig-h-chp-11-1-note-docx.pdf

Problems

1. A circle is tangent to the line $y = -4$ and passes through the point $(0, 4)$. As the circle's size and position vary, its center traces out a curve with the equation $y = ax^2$. Find the exact value of a.

2. A circle is constructed such that the latus rectum of the parabola $y^2 = 28x$ serves as its diameter. Find the (x, y) coordinates of the point(s) where this circle intersects the parabola's directrix.

3. A line with a positive slope passes through the origin and intersects the parabola $x^2 = 8y$ at a second point P. If the distance from P to the parabola's focus is exactly 10, what is the slope of the line?

4. A circle is centered at the focus of the parabola $y = \frac{1}{12}x^2$ and is tangent to its directrix. What is the straight-line distance between the two points where this circle intersects the parabola?

5. A line segment passes through the focus of the parabola $x^2 = 24y$, with both of its endpoints lying on the parabola. If one endpoint is exactly 18 units away from the parabola's directrix, what is the exact distance from the other endpoint to the directrix?

6. The parabola $y^2 = cx$ (where $c > 0$) and the circle $x^2 + y^2 = 80$ intersect at exactly two points. If the straight-line distance between these two

intersection points is equal to the focal diameter of the parabola, what is the value of c?